

Homework 3

BSTA 550

Directions

Please turn in this homework on Sakai. Please submit your homework in pdf format. You can type your work on your computer or submit a photo of your written work or any other method that can be turned into a pdf. The Adobe Scan phone app is an easy way to scan photos and compile into a PDF. Please let me know if you greatly prefer to submit a physical copy. We can work out another way for you to turn in homework.

You must show all of your work to receive credit.

Questions

1. **Cookies.** Consider a jar of 9 chocolate chip and 11 peanut butter cookies. You randomly select 2 cookies to eat. All possible choices are equally likely.
 - a. What is the probability that the 2 you select will both will be chocolate chip? *Calculate your answer “by hand.”*
 - b. What is the probability that at least one of your cookies will be peanut butter? *Calculate your answer “by hand” then verify your answer by simulating the experiment in R at least 1000 times.*
 - c. What is the probability that last 2 cookies left in the jar (after 18 have been eaten) will be chocolate chip? (Is this answer the same or different than part a? Why or why not?) *Calculate your answer “by hand” then verify your answer by simulating the experiment in R at least 1000 times.*
2. **Raffle tickets.** There are 30 raffle tickets in a bowl. Three winning tickets will be selected. Each ticket can win at most one prize. How many ways can the prizes be distributed if the following additional information is known?
 - a. All 3 winners receive goldfish (the goldfish are indistinguishable).
 - b. The 1st winner receives a car, the 2nd a bicycle, and the 3rd a goldfish.

3. A new drug is packaged to contain 30 pills in a bottle. Suppose that 98% of all bottles contain no defective pills, 1.5% contain one defective pill, and 0.5% contain two defective pills. Two pills from a bottle are randomly selected and tested. What is the probability that there are 2 defective pills in the bottle given that one of the two tested pills is defective?

Extra Problems

1. Read the Washington Post article *The amazing woman who can smell Parkinson's disease - before symptoms appear* (<http://www.washingtonpost.com/news/morning-mix/wp/2015/10/23/scottish-woman-detects-a-musky-smell-that-could-radically-improve-how-parkinsons-disease-is-diagnosed/>)

Assuming Joy Milne does not have the ability to detect Parkinson's disease via smell, answer the following questions:

- a. What is the probability of her correctly detecting Parkinson's by smelling one t-shirt?
- b. What is the probability of her correctly detecting Parkinson's in 12 out of 12 t-shirts?

2. Calculus Review

a.

$$\int_0^y c(x+y)dx$$

b.

$$\frac{d}{dx} \left(\frac{4}{9}x^2y^2 + \frac{5}{9}xy^4 \right)$$

c.

$$\frac{d}{dy} \left(\frac{4}{9}x^2y^2 + \frac{5}{9}xy^4 \right)$$

d.

$$\int_0^y 2e^{-x}e^{-y}dx$$

e.

$$\int_0^\infty xye^{-(x+y)}dy$$

f.

$$\int_x^{2x} 2e^{-(x+3y)} dy$$

- g. Find the area of the region bounded by the graphs of $f(x) = 2 - x^2$ and $g(x) = x$ by integrating with respect to x .
- h. Find the area of the region bounded by the graphs of $f(x) = 2 - x^2$ and $g(x) = x$ by integrating with respect to y .
- i. Find the area of the region bounded by the graphs of $x = 3 - y^2$ and $y = x - 1$ by integrating with respect to x .
- j. Find the area of the region bounded by the graphs of $x = 3 - y^2$ and $y = x - 1$ by integrating with respect to y .